BIOLOGY 30

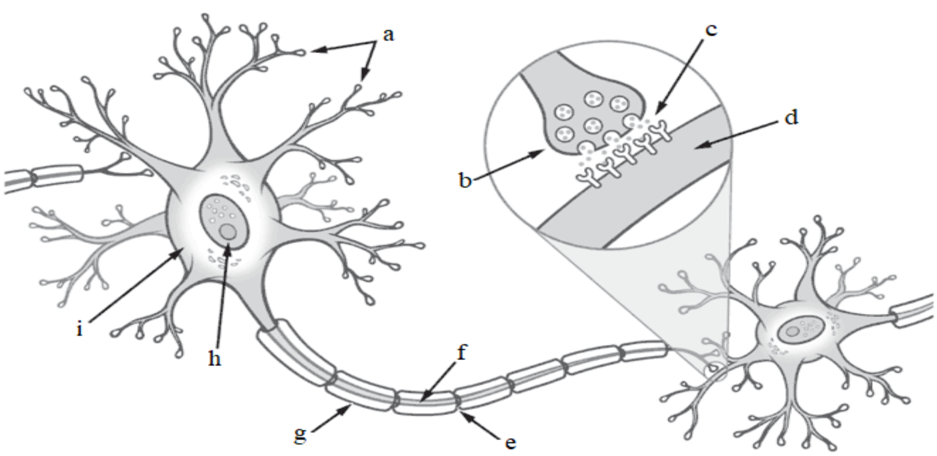
Diploma Review

UNIT 1A: THE NERVOUS SYSTEM

Topic 1: Neurons & Action Potential

- Identify the components of a neuron and describe the role of each in action potential and synaptic transmission (e.g. dendrites, axon, myelin sheath, Nodes of Ranvier, axon terminal, synapse, post-synaptic membrane).

a) Label the following diagram of a myelinated motor neuron:



b) *Explain the physiological consequences of a disorder that leads to the degeneration of the myelin sheath.*

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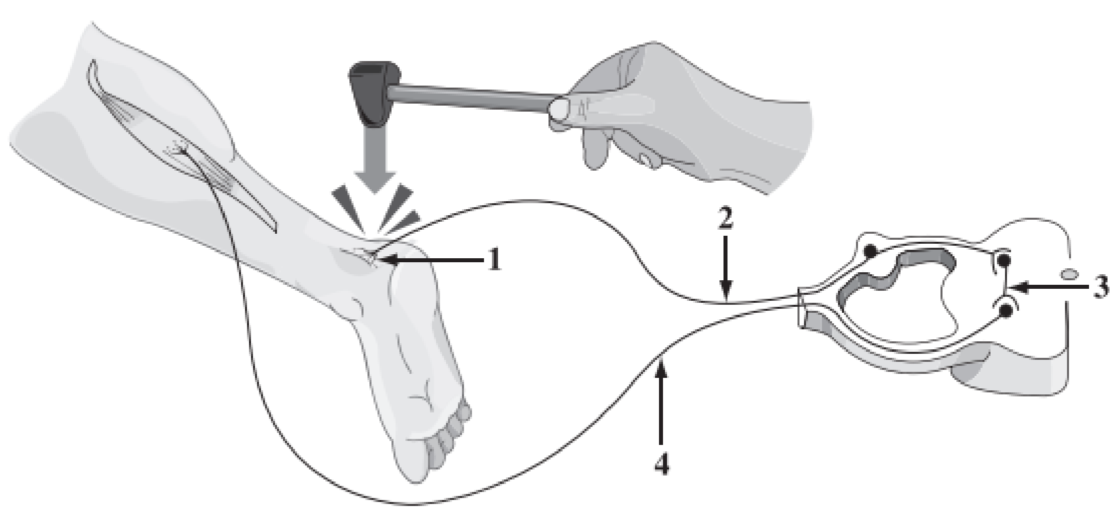
- Describe the composition and function of reflex arcs, including the roles of sensory neurons, interneurons, and motor neurons.

a) *Complete the following table.*

Type of Neuron	Function	Myelinated or Unmyelinated?
Sensory (afferent)		
Interneuron		
Motor (efferent)		

b) *Label the following diagram of a reflex arc.*

The Achilles Tendon Reflex Pathway



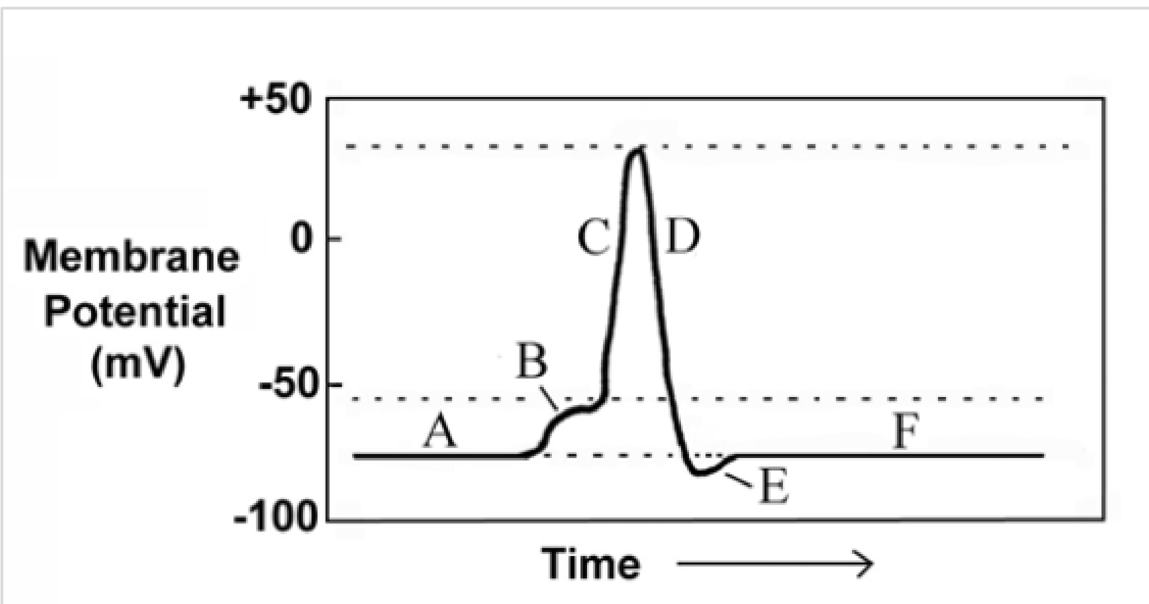
- Outline the steps involved in generating and transmitting an action potential, including the roles of sodium & potassium gates (passive transport) and sodium/potassium pumps (active transport).

a) *Determine whether each of the following statements applies to polarization, depolarization, repolarization, or the refractory period.*

	The inside of the cell is negatively charged relative to the outside, with a membrane potential of -70 mV.
	Na^+ and K^+ channels are both closed, but K^+ channels "leak", causing some K^+ to diffuse out of the cell.
	A high concentration of K^+ is found outside the cell, while a high concentration of Na^+ is found inside the cell.
	Na^+ flows into the cell through diffusion.
	The inside of the cell becomes positively charged relative to the outside, with a membrane potential of +40 mV.
	The cell is said to be "at rest".
	The cell membrane is briefly impermeable to Na^+ until resting potential is achieved.

	A high concentration of K^+ is found inside the cell, while a high concentration of Na^+ is found outside the cell.
	A stimulus of at least -55 mV is received, causing Na^+ channels to open.
	The cell is said to have reached “action potential”.
	Resting potential is restored due to the action of sodium/potassium pumps.
	The cell is said to be “hyperpolarized”
	Na^+ channels close and K^+ channels open, causing K^+ to flow out of the cell via diffusion.

- Interpret graphical representations of action potential and explain what is meant by the “all-or-none” response of neurons to stimuli of differing intensities.



a) *Identify the stage of action potential represented by each period of time in the diagram above.*

➢ PERIOD A =

- ➢ PERIOD B =
- ➢ PERIOD C =
- ➢ PERIOD D =
- ➢ PERIOD E =
- ➢ PERIOD F =

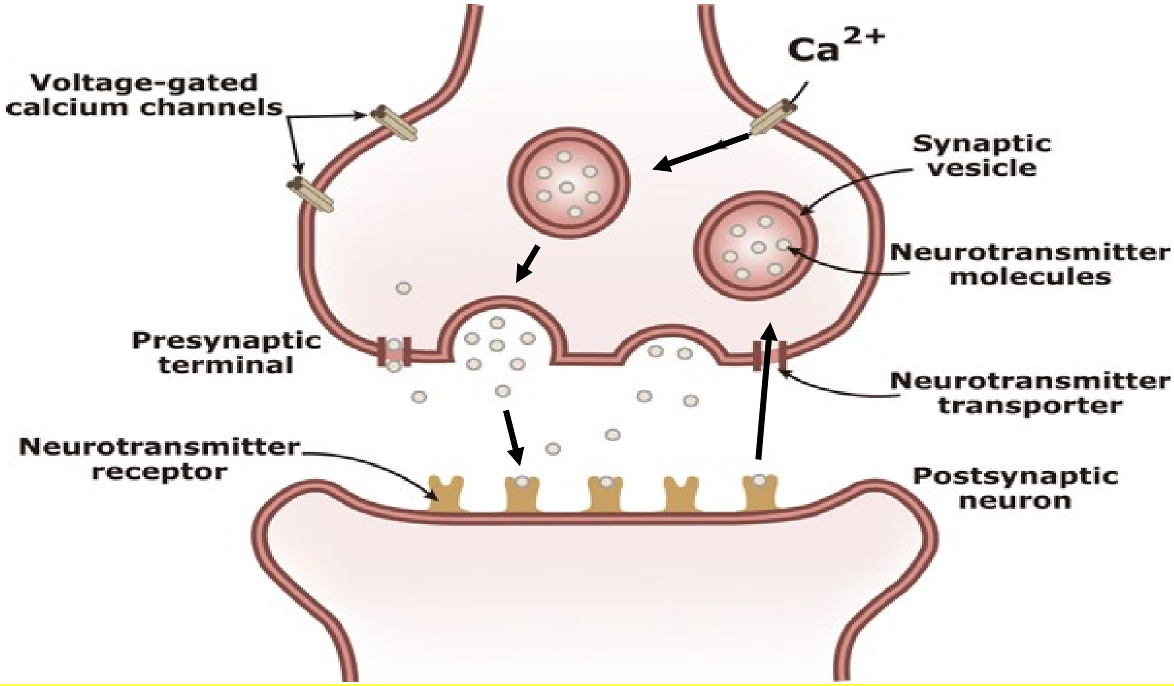
b) *Explain what is meant by the "all-or-none" response of neurons to stimuli.*

- Outline the steps involved in the transmission of a signal across a synapse.

a) *Place the following events in order with regards to synaptic transmission.*

	Ca^{2+} enters the neuron and signals neurotransmitter vesicles to move towards the presynaptic membrane.
	Neurotransmitters are taken back up by the presynaptic membrane or broken down by enzymes in the synapse.
	Action potential arrives at the axon terminal, causing voltage-gated Ca^{2+} channels to open.
	Vesicles bind to the presynaptic membrane and release neurotransmitters into the synapse via exocytosis.
	Neurotransmitters bind to receptors on the postsynaptic membrane, initiating a signal in the adjoining neuron.

b) *Label the number that corresponds with each event described in part a) on the diagram below.*



- Describe the roles of some common neurotransmitters, including norepinephrine, acetylcholine and cholinesterase.

a) Complete the following table.

Neuro-transmitter	Function	Consequences of over-production	Consequences of under-production
Nor-epinephrine			
Acetylcholine			
Cholinesterase			

- Explain the difference between excitatory and inhibitory neurotransmitters in terms of their effect on sodium/potassium gates, resting potential and threshold level required for depolarization.

  a) *Determine whether each of the following statements applies to excitatory or inhibitory neurotransmitters.*

     ➢ Depolarizes membranes by increasing permeability to Na^+

     ➢ Hyperpolarizes membranes by increasing permeability to K^+ or Cl^-

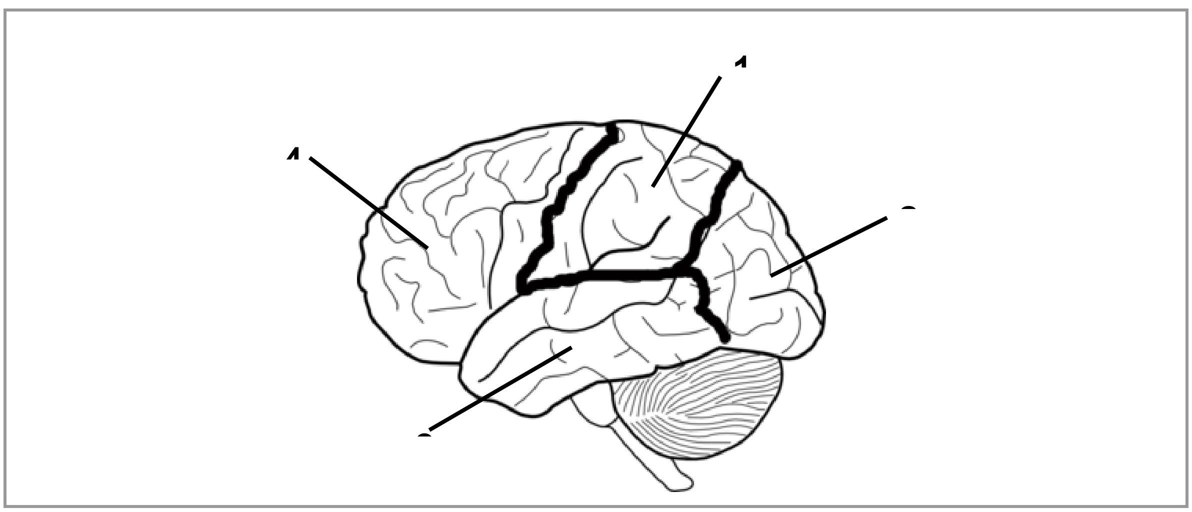
     ➢ Moves membrane potential away from the threshold (more negative)

     ➢ Moves membrane potential towards the threshold (more positive)

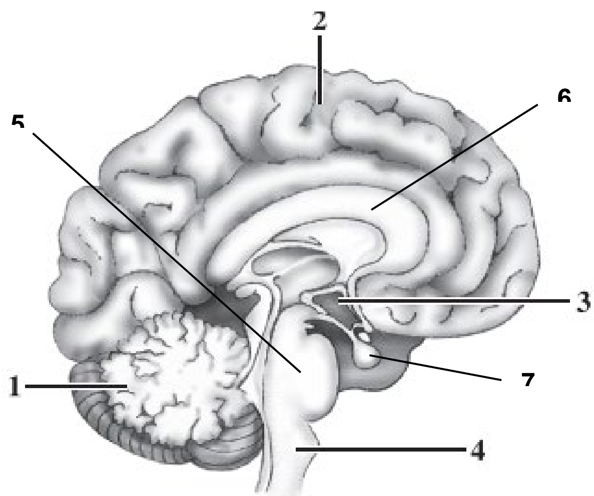
Topic 2: Central & Peripheral Nervous Systems

- Identify the principal structures of the central nervous system. Outline the roles of the four lobes of the brain (temporal, parietal, frontal, occipital) as well as other key brain structures (cerebrum, cerebellum, pons, medulla oblongata, hypothalamus, **c**orpus callosum).

  a) *Label the lobes of the brain on the diagram below. Briefly summarize the role of each lobe.*

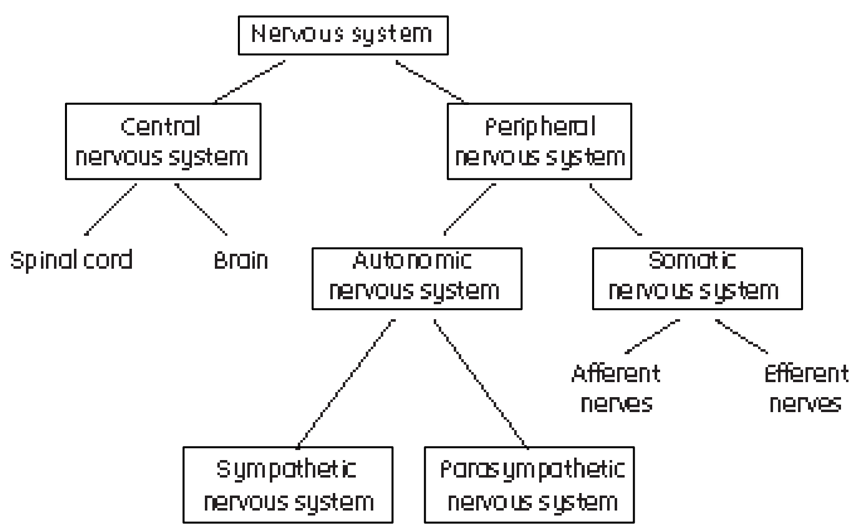


b) *Label the brain structures identified on the diagram below. Briefly summarize the role of each.*



- Explain the role of the peripheral nervous system in regulating voluntary (somatic) and involuntary (autonomic) systems. Contrast the roles of the sympathetic and parasympathetic divisions of the autonomic system.

a) *Consider the following flow-chart..*



➢ *Outline the difference between the central & peripheral nervous system.*

➢ *Outline the difference between the autonomic & somatic nervous system.*

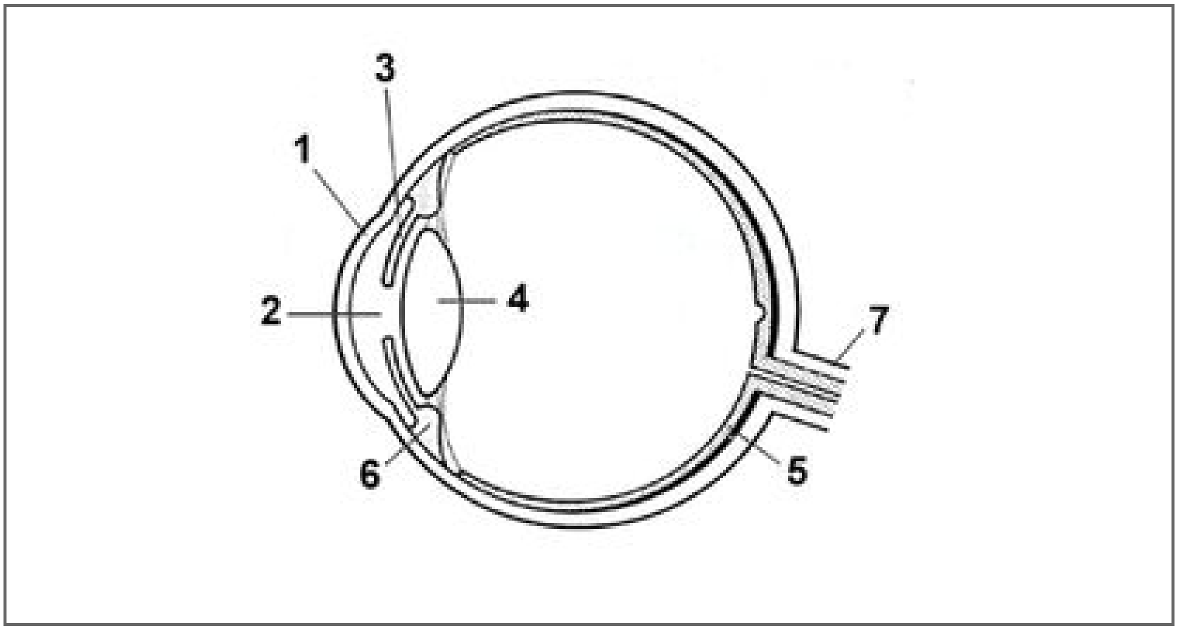
➢ *Outline the difference between the sympathetic & parasympathetic nerves.*

➢ *Outline the difference between the afferent & efferent nerves.*

Topic 3: Sensory Reception

- Identify key structures of the human eye and describe their role in photoreception (e.g. the cornea, lens, sclera, choroid layer, retina, fovea centralis, pupil, iris and optic nerve).

  a) *Label and annotate the following diagram of a human eye.*



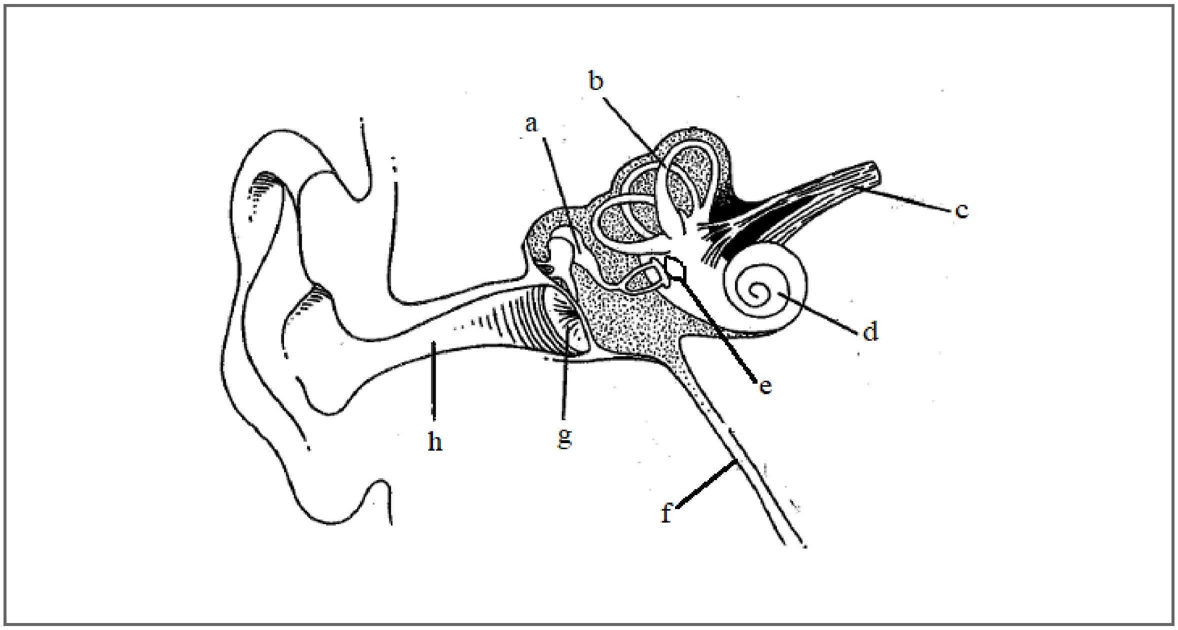
- Compare and contrast the role of rods and cones in photoreception.

  a) *Outline the physiological consequences of damage to the cones in the retina.*

  b) *Outline the physiological consequences of damage to the rods in the retina.*

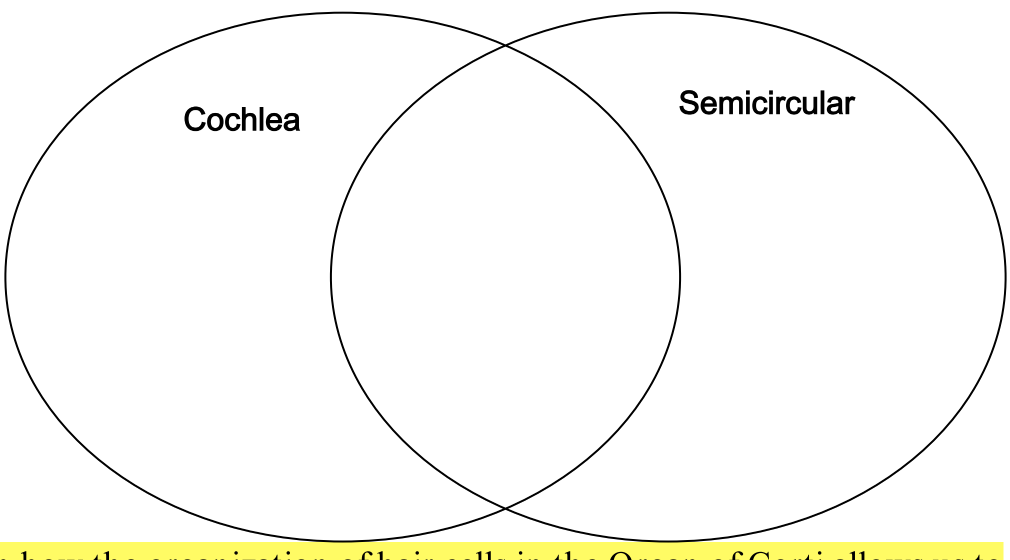
- Identify key structures of the human ear and describe their role in mechanoreception (e.g. the pinna, auditory canal, tympanum, ossicles, cochlea, organ of Corti, auditory nerve, semicircular canals and Eustachian tube).

  a) *Label and annotate the following diagram of a human ear.*



- Compare and contrast the role of the cochlea and semicircular canals in mechanoreception.

  a) *Complete the following Venn diagram comparing the cochlea and semicircular canals of the ear (list two similarities and two differences)...*



- Explain how the organization of hair cells in the Organ of Corti allows us to identify a range of sounds.

a) *What is the Organ of Corti? Explain where it is found as well as the role it plays in mechanoreception.*

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b) *Explain why some individuals with hearing loss are still able to identify low pitched frequencies of sound.*

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- Explain the roles of other sensory receptors (e.g. olfactory receptors, chemoreceptors in the nose and on the tongue, temperature and pressure receptors in the skin).

a) *What are olfactory receptors? Explain where they are found as well as the area of the brain in which their signals are interpreted.*

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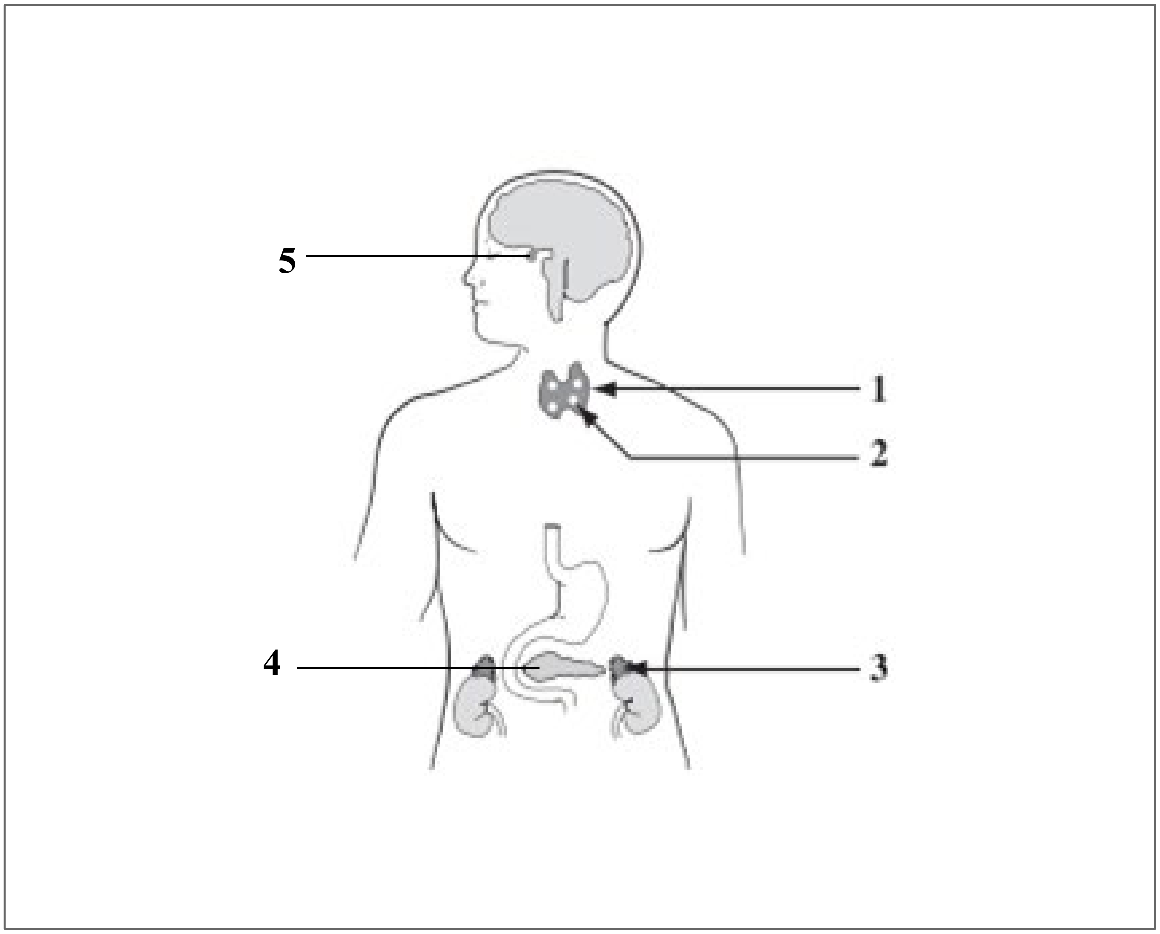
UNIT 1B: THE ENDOCRINE SYSTEM

Topic 1: Endocrine Glands

- Identify the principal endocrine glands in humans (e.g. the hypothalamus/pituitary complex, thyroid & parathyroid glands, adrenal glands, and islet cells of the pancreas).

  a) *Label the endocrine glands identified below. List the key hormones produced by each.*



Topic 2: Hormones

- Describe the roles of key hormones, including how they maintain homeostasis through negative feedback and consequences of hyper/hyposecretion (TSH &

thyroxine, calcitonin & PTH, hGH, ACTH & cortisol, glucagon & insulin, ADH, epinephrine and aldosterone).

a) *Complete the following table comparing the roles of each hormone.*

Hormone	Where it's secreted	What it targets/What it's role is
hGH		
thyroxine		
TSH		
calcitonin		
PTH		
epinephrine		
cortisol		
ACTH		
glucagon		
insulin		
aldosterone		
ADH		

b) *Complete the following table comparing the consequences of hormone imbalance.*

Hormone	Consequences of hypersecretion	Consequences of hyposecretion
hGH		
thyroxine		
TSH		
calcitonin		
PTH		
epinephrine		
cortisol		
ACTH		
glucagon		
insulin		
aldosterone		
ADH		

c) *What are "antagonistic" hormones? Provide an example.*

d) *What are "tropic" hormones? Provide an example.*

e) *Outline the process by which thyroxine and TSH work to regulate the metabolic rate of cells. Use a diagram to support your explanation.*

f) *Outline the process by which calcitonin and PTH work to regulate blood calcium levels. Use a diagram to support your explanation.*

g) *Outline the process by which insulin and glucagon work to regulate blood glucose levels. Use a diagram to support your explanation.*

h) *Explain why the body's response to epinephrine in the presence of stress occurs more rapidly than the body's response to cortisol.*

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i) *Compare and contrast the effects of aldosterone and ADH on water and ion balance in the bloodstream.*

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- Compare the nervous and endocrine systems and explain how they work together to maintain homeostasis.

a) *Complete the following table comparing the nervous & endocrine systems.*

	Nervous System	Endocrine System
Rapid or slow response?		
Type of signal		
Signals relayed through...		
Targets specific locations or whole body?		

- Infer the role of ADH and aldosterone in the maintenance of water & ion balance analyzing data on blood and urine composition; infer the role of insulin in the regulation of blood sugar by analyzing data on blood and urine composition.

a) *Describe the blood-glucose levels and blood-insulin levels expected to be observed in an individual with untreated type 1 diabetes mellitus.*

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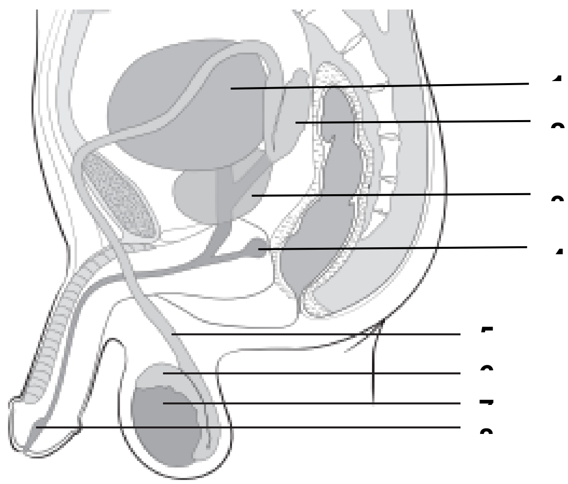
UNIT 2: REPRODUCTION & DEVELOPMENT

Topic 1: The Reproductive System

- Identify the structures of the human male reproductive system and describe their functions (e.g. testes, seminiferous tubules, epididymis, vas deferens, Cowper's gland, seminal vesicles, prostate gland, ejaculatory duct, urethra, penis)

   a) *Label the following diagram of the male reproductive system. Outline the function of each structure in the space below.*



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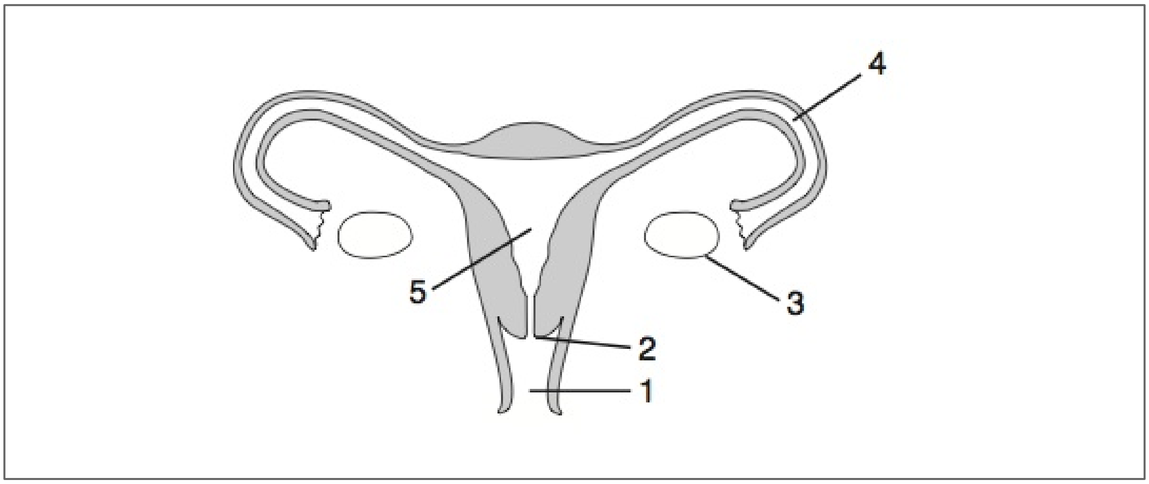
- Identify the structures of the human female reproductive system and describe their functions (e.g. ovaries, Fallopian tubes, uterus, endometrium, cervix, vagina)

a) *Label the following diagram of the female reproductive system. Outline the function of each structure in the space below.*



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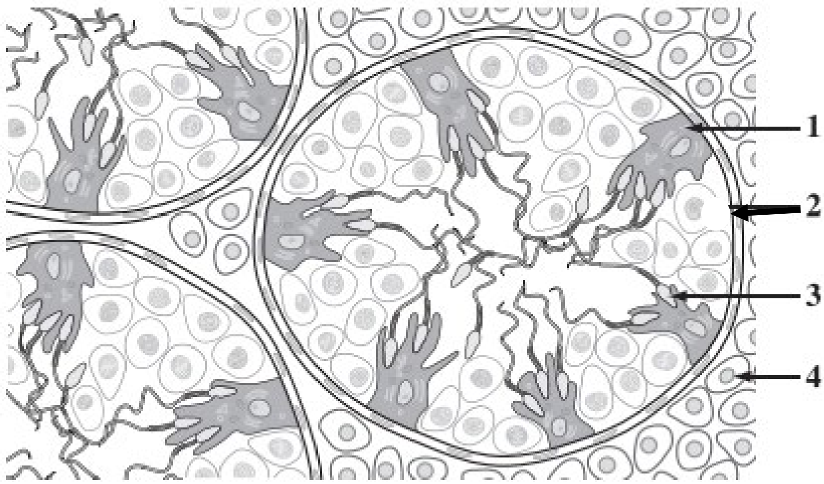
- Distinguish sperm and egg cells from their supporting structures (e.g. seminiferous tubules, interstitial cells & Sertoli cells; follicle & corpus luteum).

  a) *Identify the developing sperm cells, seminiferous tubules, Sertoli cells and interstitial cells in following cross section of a testis…*

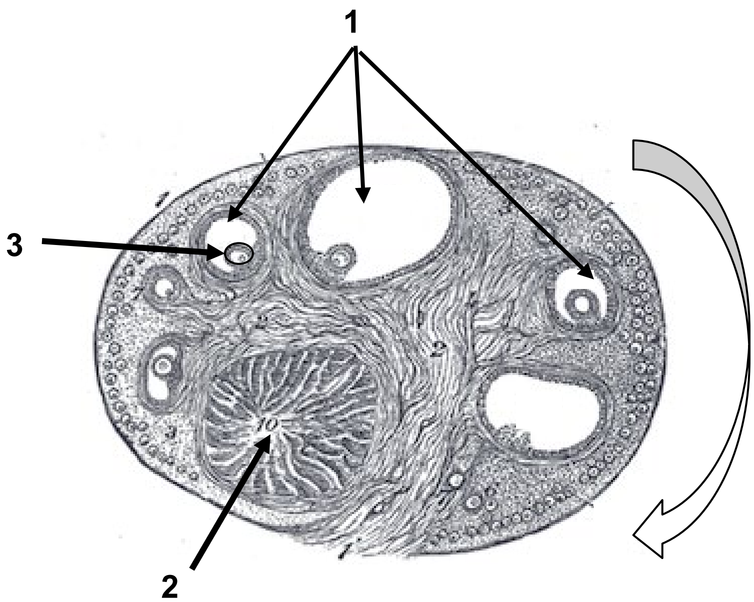
Cross Section of a Testis



  b) *Complete the following table comparing the roles of structures that support the development of sperm cells.*

Structure	Role
Seminiferous tubules	
Interstitial cells	
Sertoli cells	

c) *Identify the developing oocyte, follicles, and corpus luteum in the following cross section of an ovary…*



d) *Complete the following table comparing the roles of structures that support the development of egg cells.*

Structure	Role
Follicle	
Corpus luteum	

- Describe how sex chromosomes and hormones determine genetic and anatomic sex (i.e. the Y-chromosome and the role of testosterone)

    a) *State the anatomic sex that typically corresponds to each of the following chromosome arrangements:*

        ➢ Two x chromosomes (XX) = _______________

        ➢ One x chromosome and one y chromosome (XY) = _______________

    b) *Outline the events that occur during week 7 of development which determine whether an embryo will develop as an anatomic male or an anatomic female.*

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- Explain how STI's can interfere with fertility and reproduction.

    a) *Some STI's can lead to infertility due to the blockage of the fallopian tubes in females. Explain why this can cause difficulty conceiving.*

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- Evaluate practical solutions to decreased fertility (i.e. low sperm count, difficulty ovulating, hormone imbalance).

    a) *Low levels of this hormone may account for the inability to ovulate:*

    b) *High levels of this hormone may account for the inability to ovulate:*

Topic 2: Hormonal Regulation

- Describe the roles of hormones (GnRH, FSH, LH, estrogen, progesterone, and testosterone) in the regulation of primary and secondary sex characteristics in males and females.

a) *Complete the following table comparing primary and secondary sex characteristics in males and females.*

	Primary Sex Characteristics	**Secondary Sex Characteristics**
Definition		
Examples in males & hormones responsible		
Examples in females & hormones responsible		

b) *Outline the role of GnRH in the male & female reproductive systems.*

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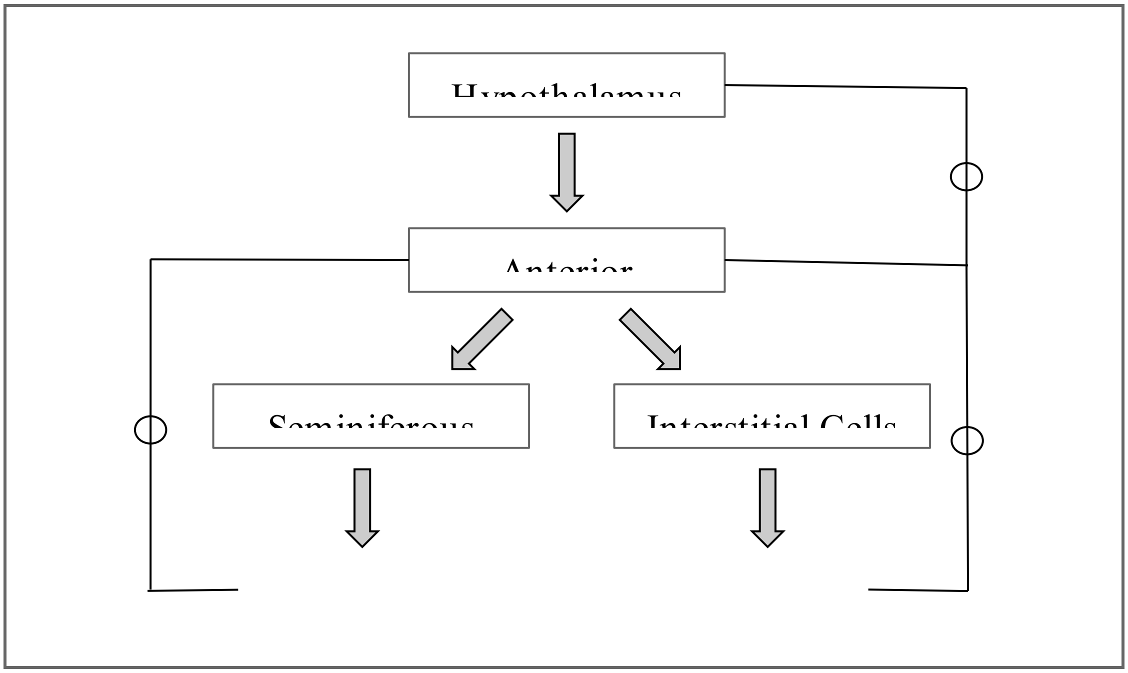
- Explain the roles of testosterone, FSH and LH in the maintenance of the male reproductive system.

a) *Complete the following flow-chart outlining the hormones involved in maintaining the male reproductive system as well as the consequences of hormone activity.*



Hormone X	Hormone Y	Hormone Z

➢ Consequence A =

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➢ Consequence B =

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- Explain the roles of estrogen, progesterone, LH and FSH in the maintenance of the menstrual cycle. Graph the changes in these hormones throughout a single menstrual cycle.

  a) *Complete the following table comparing the roles of hormones involved in the menstrual cycle.*

Hormone	Where it's secreted	What it does
FSH		
Estrogen		
LH		
Progesterone		

  b) *What is a "gonadotropic" hormone? Provide two examples.*

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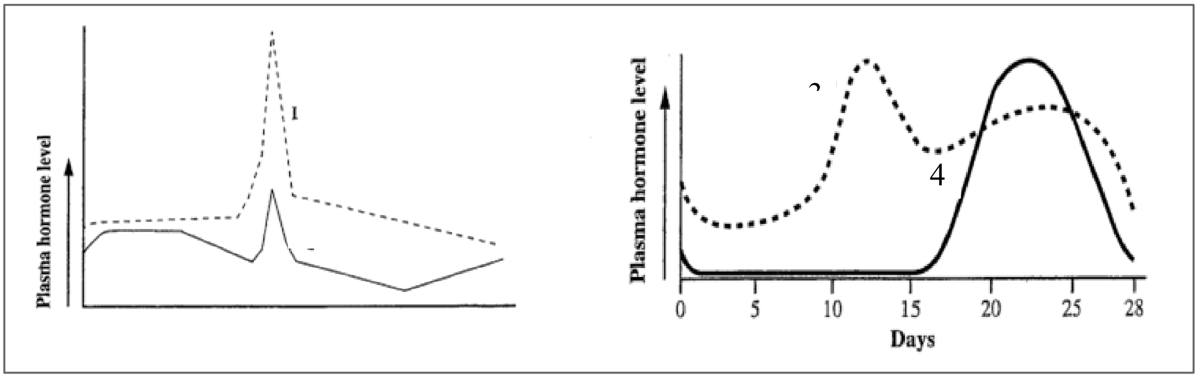
c) *Label the hormones graphed below…*



➢ Hormone 1 = ________________________________

➢ Hormone 2 = ________________________________

➢ Hormone 3 = ________________________________

➢ Hormone 4 = ________________________________

d) *Determine the stage(s) of the menstrual cycle to which each of the following statements apply:*

➢ A single follicle begins to develop within the ovary.

➢ A follicle releases a mature egg into the fallopian tube.

➢ A follicle breaks down to form the corpus luteum.

➢ The endometrial lining is shed.

- ➢ The endometrial lining begins to thicken.
- ➢ Hormone levels are at their lowest.
- ➢ Hormones are secreted to prevent further ovulation.
- ➢ Hormones are secreted to prevent the shedding of the endometrium.

Topic 3: Fertilization and Development

- Trace the processes of fertilization and implantation with regards to the main physiological events that occur during each stage of development (zygote, blastocyst, gastrulation). Describe the hormonal control mechanisms behind these events (e.g. hCG, progesterone, LH).

a) *Complete the following table comparing the various structures involved in the development of an embryo prior to implantation.*

Structure	What it is	When it develops (# of days after fertilization)	Diagram
Gamete			
Zygote			
Morula			

Blastocyst			
Gastrula			

b) *What is "cleavage"? How does it differ from regular cell division?*

c) *What is the purpose of gastrulation?*

d) *Outline the role of human chorionic gonadotropin (hCG) in the maintenance of a pregnancy.*

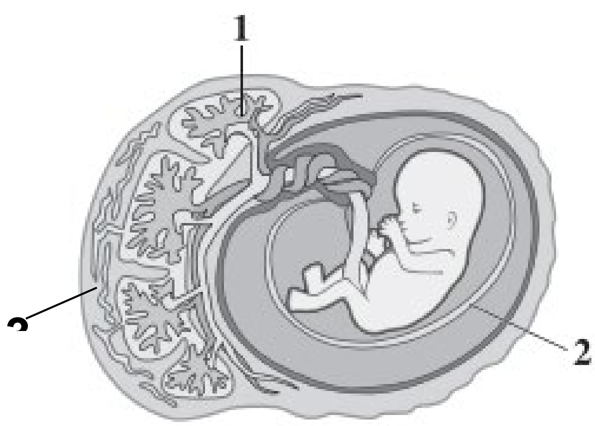
- Identify major tissues and organs that arise from the differentiation of the ectoderm, mesoderm, and endoderm in the embryo.

   a) *Complete the following table comparing the three germ layers of the gastrula.*

Germ Layer	What it develops into
Endoderm	
Mesoderm	
Ectoderm	

- Outline the extra-embryonic membranes which act to support a developing embryo/fetus (e.g. placenta, chorion, amnion, allantois).

   a) *Label the placenta, chorion, and amnion on the following diagram. Annotate their function in the space below.*



➢ Placenta:

➢ Chorion:

➢ Amion:

b) *Outline the roles of the allantois and the yolk sac. Why are they not visible in the diagram above?*

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- Describe the development of organ systems during each major stage (trimester) of pregnancy.

a) *Complete the following table comparing the first, second, and third trimesters of pregnancy.*

Trimester	Weeks	Characteristics
First		
Second		
Third		

- Describe the influence of environmental factors on embryonic and fetal development.

a) *What is a teratogen? Provide an example, and explain how it impacts fetal development.*

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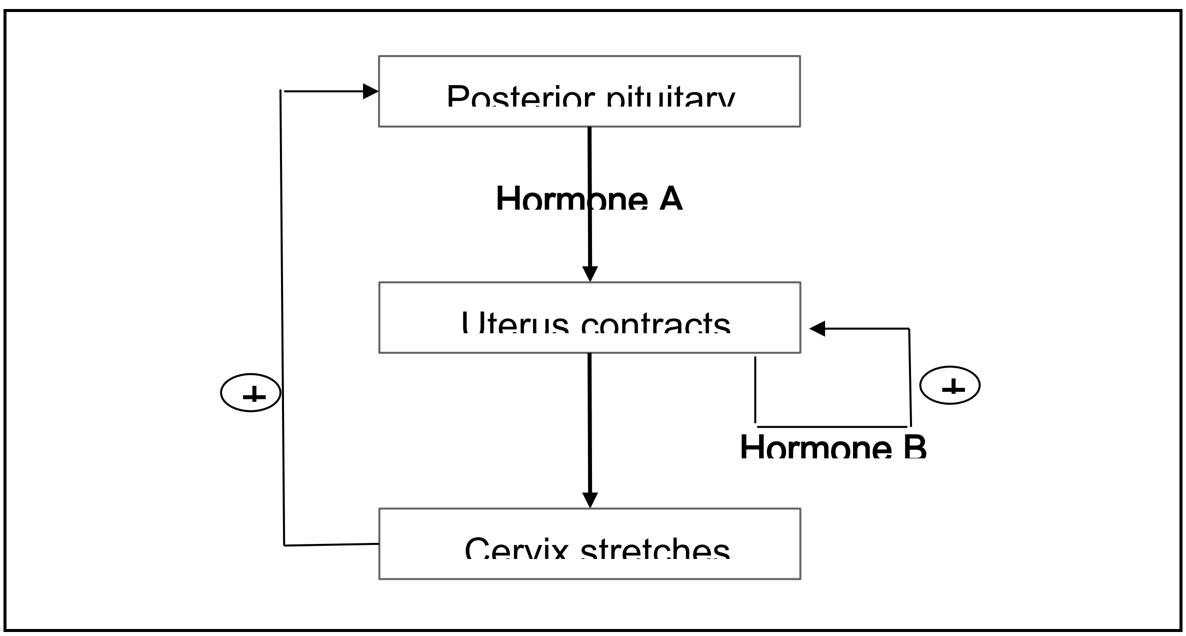
- Describe the physiological basis of various reproductive technologies (e.g. contraceptives, in vitro fertilization, fertility treatments).

  a) Outline the process of in vitro fertilization.

- Describe the processes of parturition and lactation, including the hormonal control mechanisms behind these events (e.g. prostaglandins, oxytocin, prolactin)

  a) Complete the following flow-chart outlining the process of parturition.



➢ Hormone A = ______________________

➢ Hormone B = ______________________

b) *Hormone A is also involved in the process of lactation. Outline the role of this hormone in lactation, as well as the role of the hormone prolactin.*

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UNIT 3A: CELL DIVISION

Topic 1: Mitosis

- Describe chromosome number ("ploidy") in somatic vs. sex cells.

a) *Complete the following table comparing somatic cells and sex cells.*

Type of Cell	Ploidy	Where it's found
Somatic Cell		
Sex Cell		

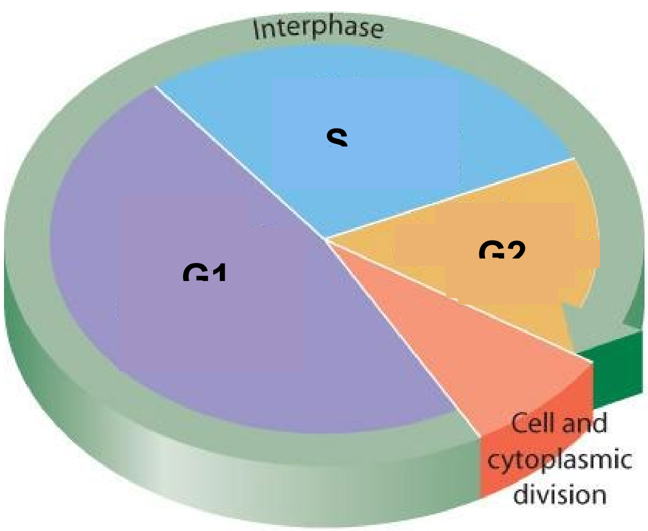
b) *Define each of the following terms.*

➢ Homologous chromosome

➢ Non-homologous chromosome

- ➢ Autosome
- ➢ Sex chromosome
- ➢ Somatic cell
- ➢ Sex cell
- ➢ Sister chromatid

- Explain the events of the cell cycle (interphase, mitosis, cytokinesis) and use microscope images to calculate the duration of each stage (mitotic index).



a) Outline the key events that occur during interphase.

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b) *Which phase do healthy cells spend most of their life cycles in? Explain why.*

- Outline the major events that occur during each phase of mitosis. Identify diagrams of cells undergoing each phase of mitosis.

a) *Outline the major events that occur during each of the following phases of mitosis. Sketch a diagram of a cell undergoing each stage.*

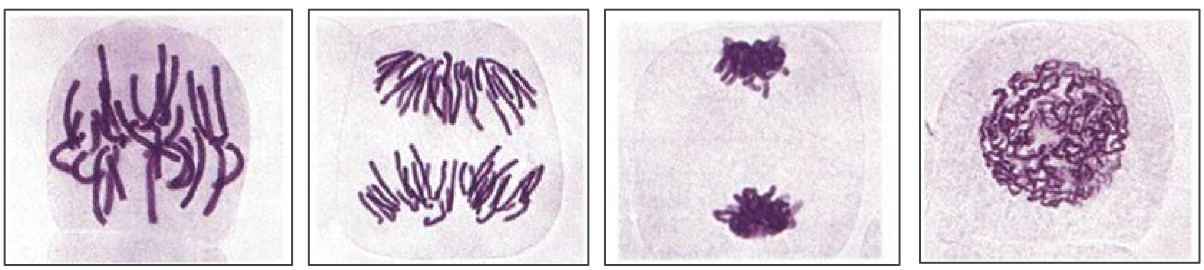
Prophase

Metaphase

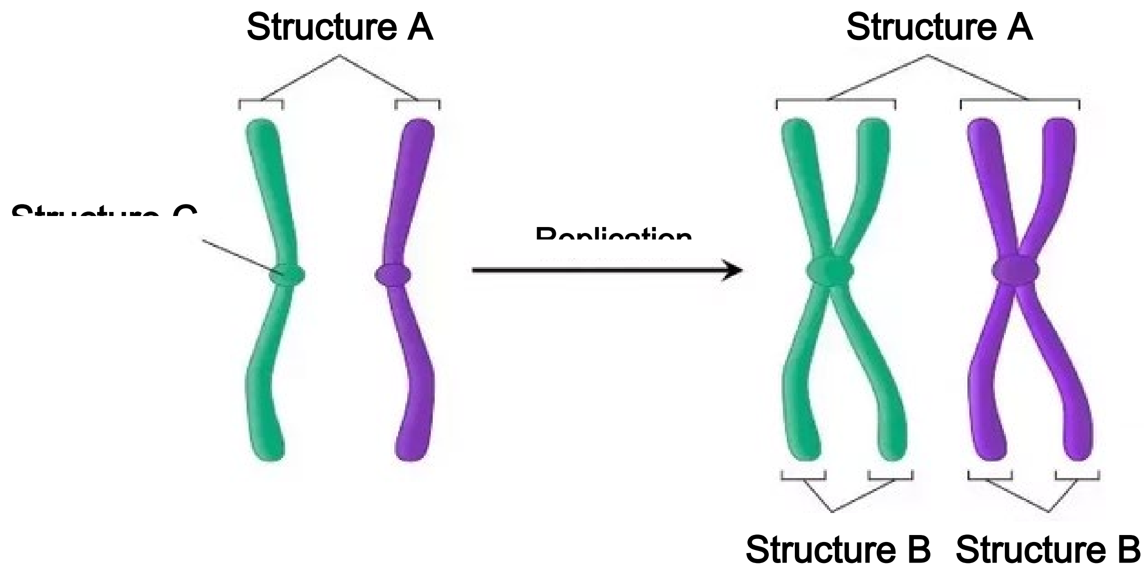
Anaphase

Telophase

b) Identify the stage of mitosis each of the following cells is undergoing:



c) Identify each structure labeled below:



➢ Structure A = ______________________________

➢ Structure B = ______________________________

➢ Structure C = ______________________________

Topic 2: Meiosis

- Describe the necessity for the reduction of chromosome number during spermatogenesis and oogenesis.

  a) *Outline the main purpose of meiosis.*

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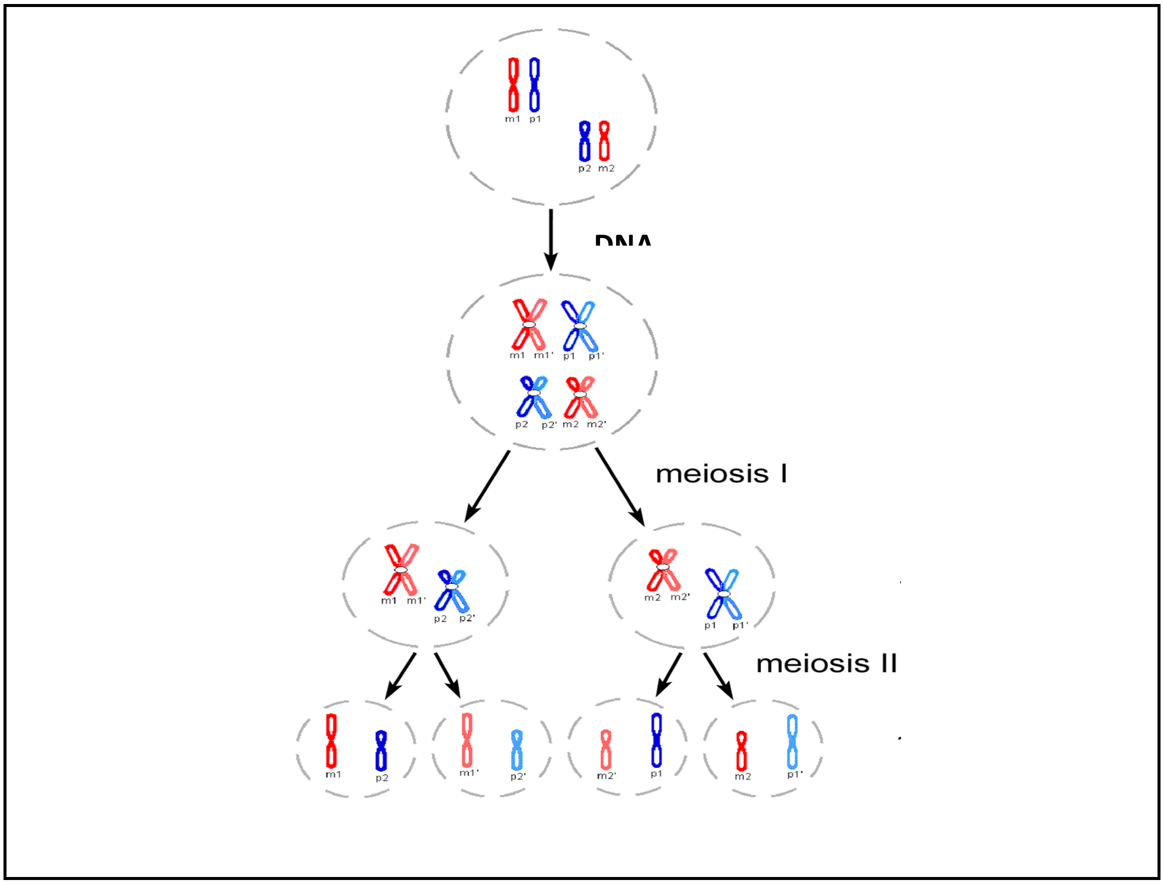
- Outline the major events that occur during each phase of meiosis, including the significance of crossing over and independent assortment. Identify diagrams of cells undergoing each phase of meiosis.

a) *Complete the following table comparing the mechanisms by which meiosis increases the genetic diversity of gametes.*

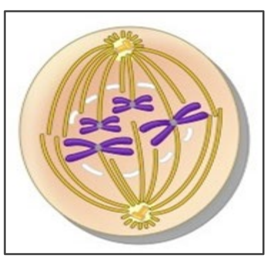
	Crossing Over	Independent Assortment
When it occurs		
How it increases diversity		
Diagram of cell during that stage		

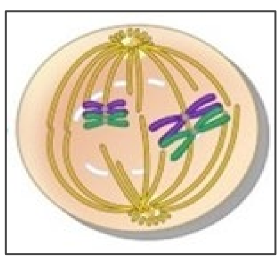
b) *Label the ploidy of each cell below prior to DNA replication, after DNA replication, after the first meiotic division, and after the second meiotic division. At what point are the cells considered to be "gametes"?*

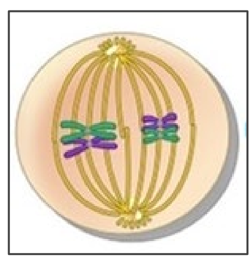


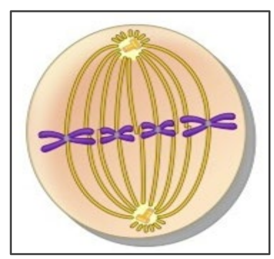
- Compare and contrast the processes of mitosis and meiosis.

a) Identify the stage of mitosis/meiosis each of the following images applies to:

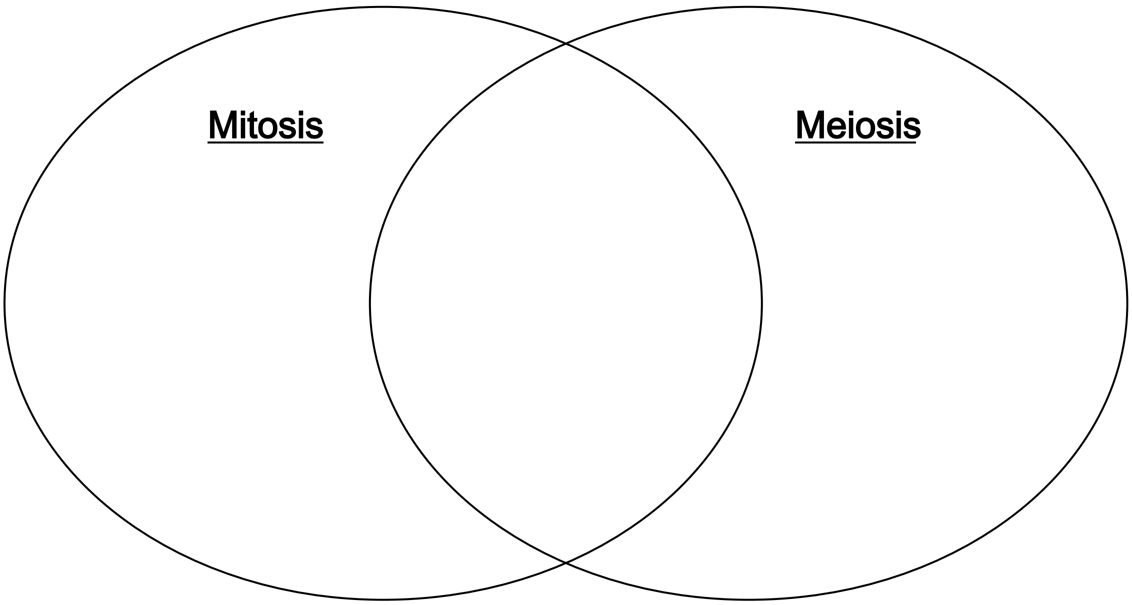








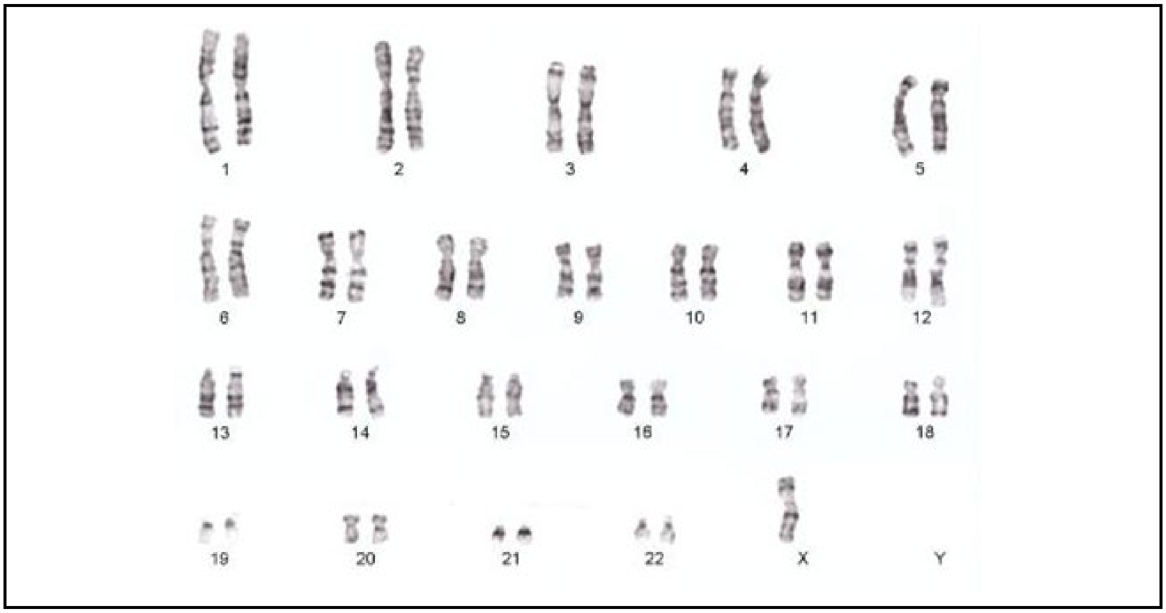
b) *Complete the following Venn diagram comparing mitosis & meiosis:*



- Interpret human karyotypes and describe the impact of nondisjunction on chromosome number.

a) *Using a diagram, illustrate how nondisjunction can lead to the formation of gametes with too many or too few chromosomes.*

b) Consider the following karyotype…



➢ How many pairs of homologous chromosomes does this person have?

➢ What is atypical about this person's karyotype?

➢ What is the most likely anatomic sex of this person?

- Compare the formation of fraternal (dizygotic) and identical (monozygotic) twins.

a) Explain how the formation of fraternal twins differs from the formation of identical twins.

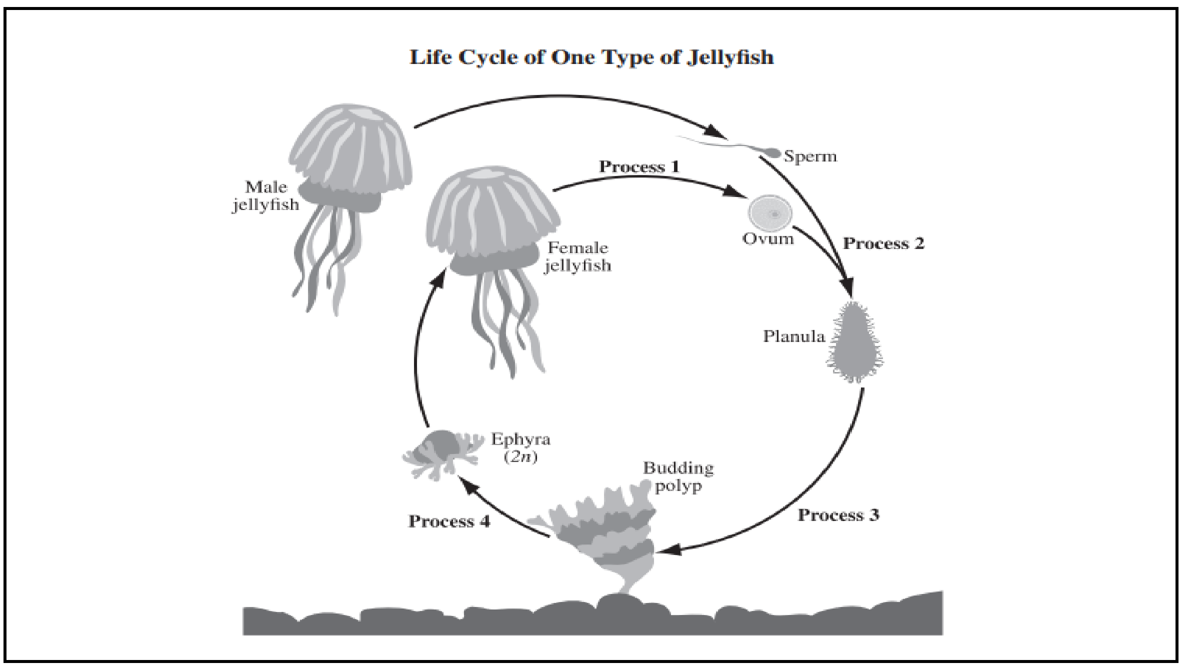
Topic 3: Alternation of Generations

- Compare a range of reproductive strategies (e.g. binary fission, budding, sexual reproduction).

   a) *Outline some key differences between sexual and asexual reproduction.*

- Describe the diversity of reproductive strategies by comparing the alternation of generations in a range of organisms.

   a) *Label the following diagram of a jellyfish life cycle:*



➢ Process 1 = ________________________________

➢ Process 2 = ________________________________

➢ Process 3 = ________________________________

➢ Process 4 = ________________________________

UNIT 3B: MENDELIAN GENETICS

Topic 1: Monohybrid Crosses

- Describe the evidence for dominance, segregation and the independent assortment of genes on different chromosomes, as investigated by Mendel.

a) *Define each of the following terms. Provide an example of each.*

Term	Definition	Example
Gene		
Allele		
Genotype		
Phenotype		

b) *Outline Mendel's Law of Segregation. What evidence did he use to come to this conclusion?*

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c) *Outline Mendel's Law of Independent Assortment. What evidence did he use to come to this conclusion?*

- Compare ratios and probabilities of genotypes and phenotypes for dominant and recessive alleles using monohybrid Punnett Squares.

a) *Hyperelastosis cutis (HC) is an autosomal recessive disorder in horses that causes the skin to tear easily. If an unaffected female horse whose father had HC is mated with a male horse who has HC, determine the probability that their offspring will also have HC.*

b) *Marfan's Syndrome is an autosomal dominant disorder in humans that affects the connective tissue of limbs. Is it possible for an unaffected man and woman to produce a child with Marfan's Syndrome? Explain.*

Topic 2: Dihybrid Crosses

- Compare ratios and probabilities of genotypes and phenotypes for dominant and recessive alleles using dihybrid Punnett Squares.

a) *In garden peas, the allele for tall plant height (T) is dominant over the allele for short plant height (t), and the allele for axial flower position (A) is dominant over the allele for terminal flower position (a)..*

A plant that is homozygous dominant for both the height and flower position trait is crossed with a plant that is heterozygous for the height trait and homozygous recessive for the flower position trait…

1. Determine the phenotype of each plant described above.

2. Determine the genotypes of the gametes produced by each plant in this cross.

3. Determine the phenotypic ratio of offspring produced by the cross described above.

Topic 3: Other Modes of Inheritance

- Compare ratios and probabilities of genotypes and phenotypes for multiple, incompletely dominant, and codominant alleles.

a) *Complete the following table comparing incomplete and codominance:*

	Definition	Phenotype of heterozygotes
Incomplete Dominance		
Codominance		

b) *Tay-Sachs disease is an autosomal recessive disorder which demonstrates incomplete dominance. As a result, heterozygotes are only mildly affected by the disease, whereas individuals who possess two recessive alleles rarely live past the age of ten.*

A man who is heterozygous for Tay-Sachs disease has a child with a woman who does not have Tay-Sachs…

1. Determine the probability that the child will be a girl with the severe form of Tay-Sachs.

2. Determine the probability that the child will be a girl with the mild form of Tay-Sachs.

c) *In humans, blood type is determined by multiple alleles. The alleles "I^A" and "I^B" are codominant. They are also both dominant over the allele "i". The following phenotypes may occur as a result:*

Phenotype (Blood Type)	Possible Genotypes
Type A	$I^A I^A$ *or* $I^A i$
Type B	$I^B I^B$ *or* $I^B i$
Type AB	$I^A I^A$
Type O	ii

- ➢ A woman with Type A blood and a man with Type B blood have a child with Type O blood. What must be their genotypes?

- ➢ A heterozygous woman with Type A blood has a child with a man with Type AB blood. What is the probability that the child will have Type B blood? Use a Punnett Square to support your answer.

- Explain the relationship between variability and the number of genes controlling a given trait.

  a) *Outline the difference between polygenic and pleiotropic traits. Provide an example of each.*

- Compare the pattern of inheritance produced by genes on the sex chromosomes to that produced by genes on autosomes, as investigated by Morgan.

  a) *Colour-blindness demonstrates an X-linked recessive mode of inheritance in humans. Complete the following table summarizing the phenotypes that correspond to each possible genotype with regards to colour-blindness…*

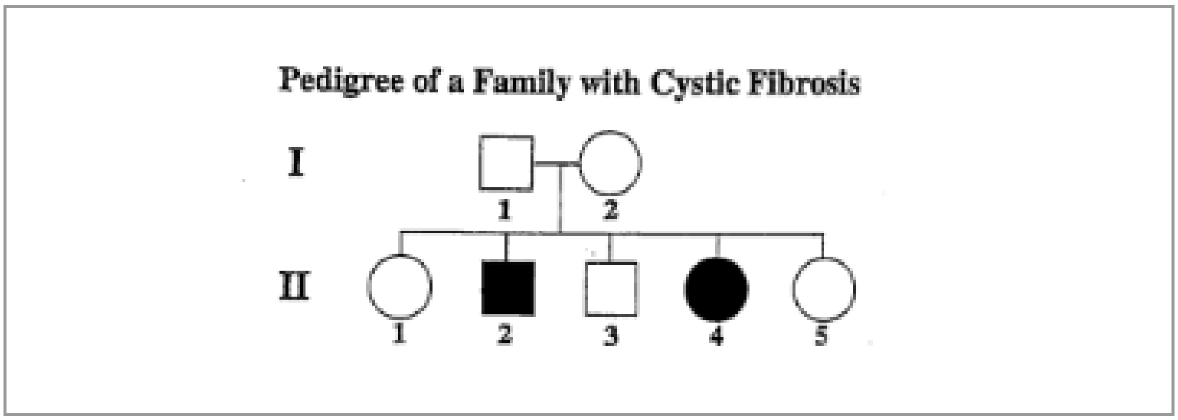
Genotype	Phenotype	Sex
X^BY		
X^bY		
X^BX^B		
X^BX^b		
X^bX^b		

b) A colour-blind man and a homozygous woman with normal vision are expecting a child What is the probability that, if the child is male, he will also be colourblind?

Topic 4: Pedigrees

- Use a pedigree to predict the genotypes of individuals as well as the probability of inheriting a trait given different modes of inheritance.

a) The following pedigree demonstrates the inheritance of cystic fibrosis. Cystic fibrosis is caused by a recessive allele that is found on chromosome 7…

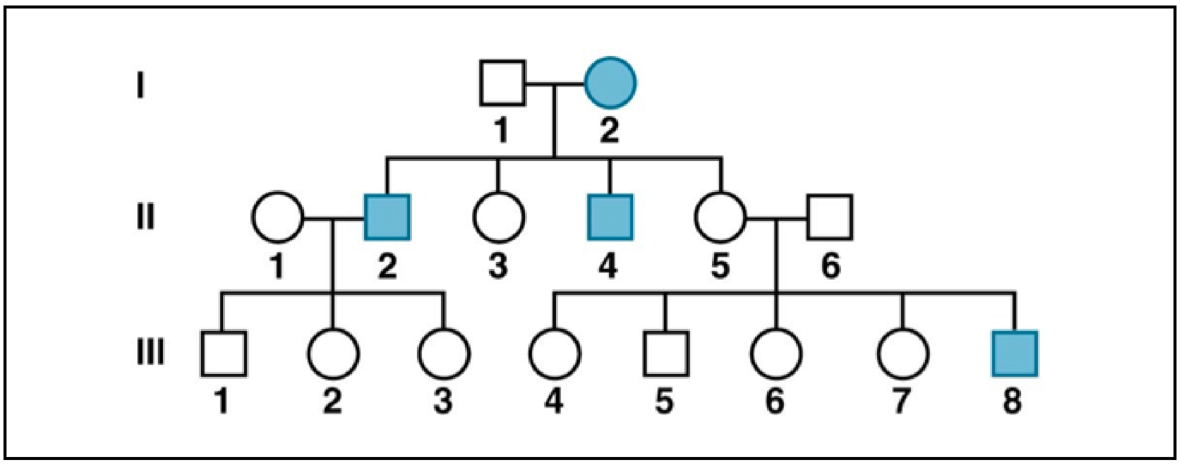


➢ Determine the genotypes of individuals I-1 and I-2.

➢ Determine the genotypes of individuals II-2 and II-4.

➢ If individuals I-1 and I-2 were to have another child, what is the probability that they would be a carrier of cystic fibrosis? Use a Punnett Square to support your answer.

b) *The following pedigree demonstrates the inheritance of an X-linked recessive trait...*



➢ If individuals I-1 and I-2 have another son, what is the probability that he would have the recessive trait?

➢ Is it possible for individuals II-1 and II-2 to have an affected son? Explain why or why not.

➢ If individuals II-5 and II-6 have another child, what is the probability that they will have the recessive trait?

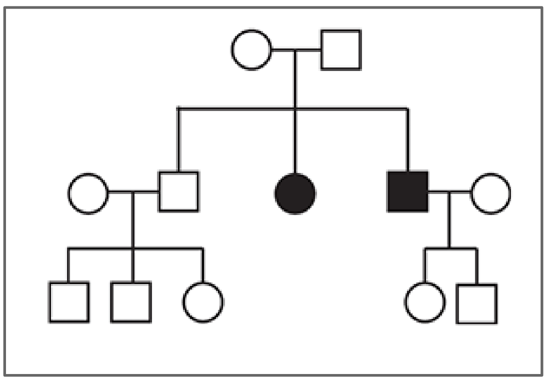
- Interpret mode of inheritance given a pedigree.

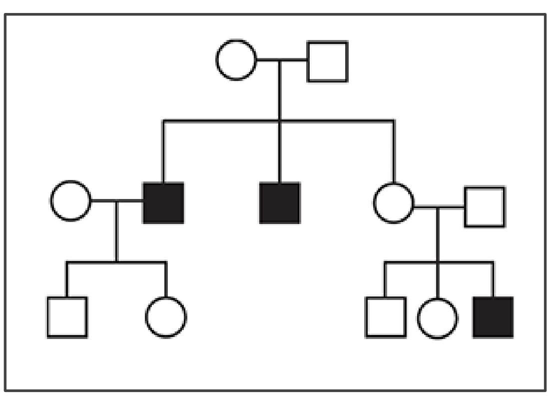
a) *Complete the following table summarizing the "clues" used to determine the mode of inheritance of a particular trait given a pedigree.*

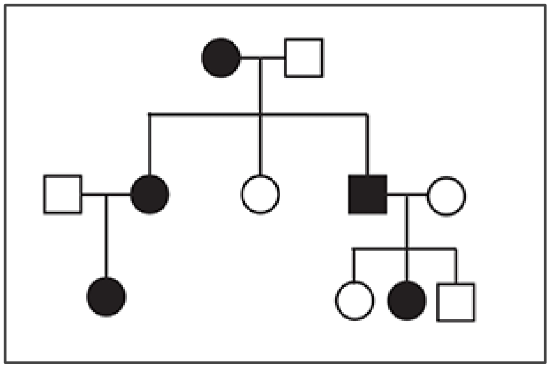
Mode of Inheritance	"Clues" to look for in a pedigree	Example pedigree
Autosomal dominant		
Autosomal recessive		
X-linked dominant		
X-linked recessive		

Y-linked		

b) *Determine the most likely mode of inheritance of each of the following pedigrees. Explain your reasoning in the space provided.*







Topic 5: Gene Linkage & Mapping

- Explain the influence of gene linkage and crossing over on variability.

  a) Does crossing over increase or decrease the genetic diversity of gametes?

  b) Does gene linkage increase or decrease the genetic diversity of gametes?

- Explain why predicted phenotypic ratios may differ from the actual counts in a dihybrid cross with regards to gene linkage.

  a) What evidence is used to determine whether or not two genes are linked?

- Analyze the crossover frequencies between genes on a single pair of chromosomes to create a chromosome map.

    a) *Create a chromosome map using the following information:*

Cross-over Frequencies of Some Genes on Human Chromosome 22

Genes	Cross-over Frequency
1 and 4	25.9%
1 and 3	18.8%
2 and 3	19.9%
2 and 4	12.8%
3 and 4	7.1%

UNIT 3C: MOLECULAR GENETICS

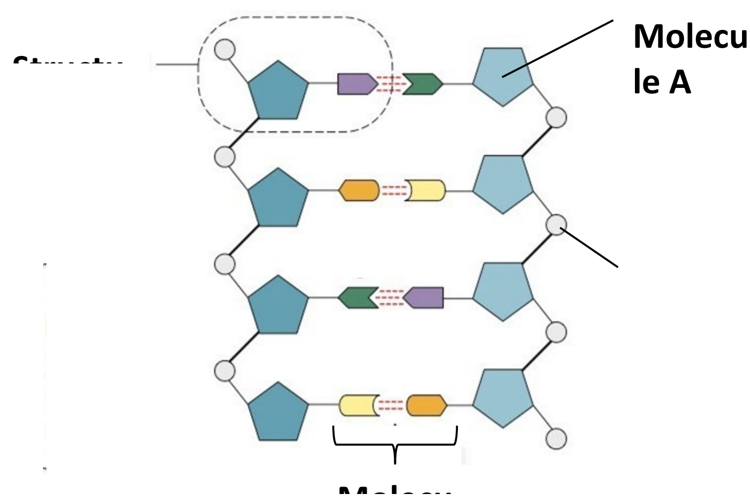
Topic 1: DNA Structure & Replication

- Summarize the historical events that led to the discovery of the structure of the DNA molecule, including the work of Franklin, Watson and Crick.

    a) *Outline how the evidence gathered by Rosalind Franklin helped Watson and Crick develop their “doublehelix” model of the DNA molecule.*

- Describe the structure of the DNA molecule, including the sugar-phosphate backbone, nitrogenous bases, and directionality.

a) Label the following diagram of a DNA molecule…



b) Complete the following doublestranded DNA sequence by adding the complementary base pairs:

A A C A G T C T G A T A T G G C A C

| | | | | | | | | | | | | | | | | |

c) What type of bonds are the nitrogenous bases in a DNA molecule held together by?

d) *The DNA molecule is said to be "antiparallel". Explain what is meant by this.*

- Explain how DNA molecules replicate themselves, including the roles of major enzymes (e.g. helicase, primase, DNA polymerase, ligase).

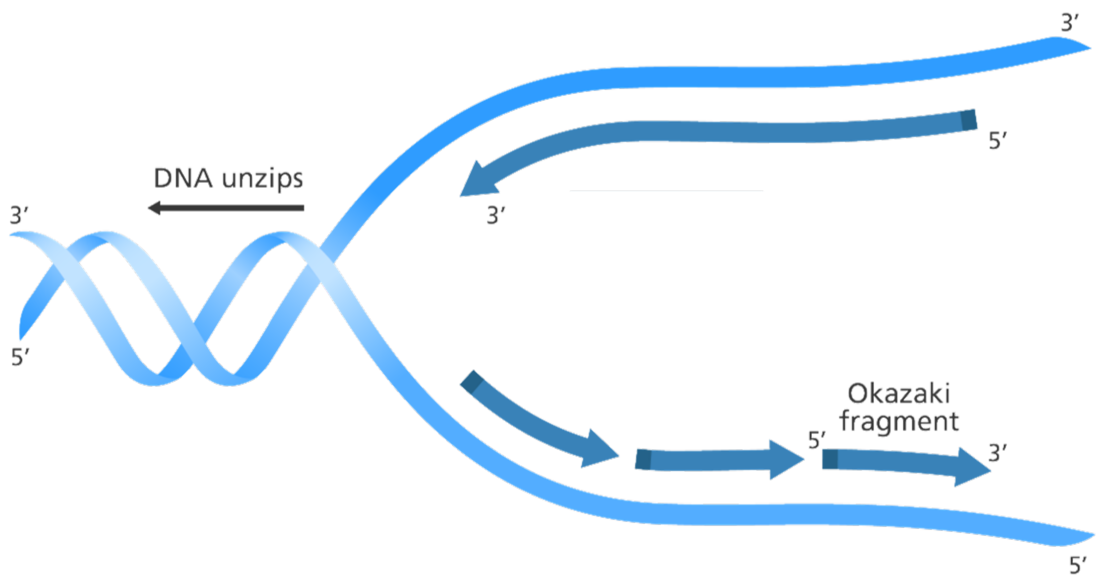
a) *The process of DNA replication is said to be "semiconservative". Explain what is meant by this.*

b) *Complete the following table comparing the enzymes involved in the process of DNA replication.*

Enzyme	Role
Helicase	
Primase	
DNA Polymerase	

Ligase	

c) *Identify the leading and lagging strand in the following diagram of DNA replication:*



Topic 2: Transcription & Translation

- Describe how genetic information is transcribed into mRNA molecules and translated into sequences of amino acids.

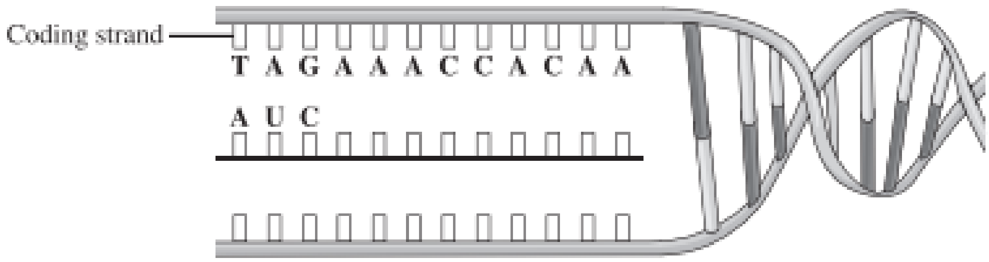
a) *Outline four differences between DNA and RNA.*

DNA	RNA

b) Determine whether each of the following statements applies to transcription or translation.

	Occurs in the cytoplasm
	Occurs in the nucleus
	Converts DNA to mRNA
	Converts mRNA to an amino acid chain
	Requires RNA polymerase
	Requires ribosomes

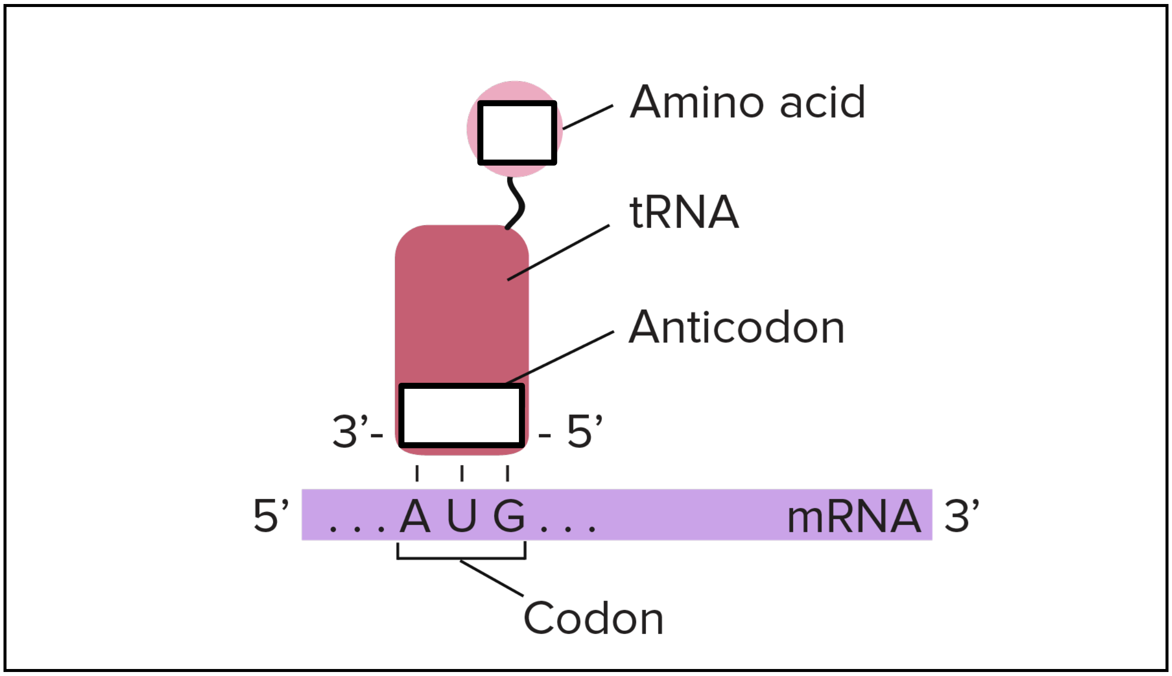
c) Transcribe the following coding strand of DNA into mRNA. Then, translate the mRNA into a sequence of amino acids in the space below:



➢ mRNA = ____

➢ amino acid sequence = ____

d) *Identify the tRNA anticodon that matches the following mRNA sequence, as well as the corresponding amino acid.*



Topic 3: Mutations & Genetic Technologies

- Explain how random changes (mutations) in the sequence of nitrogenous bases may result in abnormalities (e.g. cancer) or increase genetic variability.

a) Explain why changes to a single nitrogenous base often do not result in the production of a different amino acid sequence.

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b) Which of the following mutations will result in the production of a stop codon in the following gene?

AAA ATA TTC CAG GTC

- ❑ AAA ATA TTC CAC GTC
- ❑ AAA ATC TTC CAG GTC
- ❑ TCT ATA TTC CAG GTC

c) *Outline the difference between a frameshift mutation and a point mutation.*

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- Explain how cells are transformed by inserting new DNA sequences into their genomes. Outline the role of restriction enzymes and ligase in this process.

a) *Summarize the role of restriction enzymes and ligase in the creation of a transgenic organism.*

➢ **Restriction enzymes**

➢ **Ligase**

- Explain how DNA contained within the nucleus, mitochondria and chloroplast provide evidence for the evolutionary relationships between different species.

  a) *Explain how the inheritance of mitochondrial DNA differs from the inheritance of nuclear DNA.*

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UNIT D: POPULATION & COMMUNITY DYNAMICS

Topic 1: Hardy-Weinberg Theory

- Use the conditions of Hardy-Weinberg equilibrium to explain why populations change over time.

  a) *List the five conditions that must be met in order for a population to be at Hardy-Weinberg equilibrium:*

     1.

     2.

     3.

     4.

5.

- Use the Hardy-Weinberg principle ($p+q=1$ and $p^2+2pq+q^2=1$) to calculate allele and genotype frequencies.

    a) *What do "p" and "q" in the formula $p+q=1$ represent?*

    b) *What do "p^2", "2pq", and "q^2" in the formula $p^2+2pq+q^2=1$ represent?*

    c) *A particular gene has only two alleles, R and r. The frequency of "r" in a given population is 17%. Determine the frequencies of each genotype (RR, Rr and rr) in this population.*

    d) *In mice, brown fur is dominant to white fur. A large population of mice consists of 1100 individuals, 758 of which have brown fur. Determine the frequency of the dominant and recessive alleles for fur colour in this population of mice.*

e) *In fruit flies, eye colour is determined by two alleles. The dominant allele (R) results in red eyes, while the recessive allele (r) results in white eyes. In a population of 860 individuals, only 3% of the population has white eyes. Determine how many flies have each genotype in this population.*

Topic 2: Causes of Gene Pool Change

- Describe the factors that lead to gene pool change (e.g. natural selection, genetic drift, gene flow, nonrandom mating, bottleneck effect, founder effect, mutation).

a) *Complete the following table summarizing the factors that lead to gene pool change:*

	Example of how it contributes to gene pool change	Does it increase or decrease diversity?
Natural selection		
Genetic drift		
Gene flow		

Nonrandom mating		
Bottleneck effect		
Founder effect		
Mutation		

Topic 3: Population Growth

- Describe the influence of various species interactions on populations (e.g. predator-prey relationships, producer-consumer relationships, symbiotic relationships, interspecific and intraspecific competition).

  a) *Differentiate between interspecific and intraspecific competition. Provide an example of each to support your explanation.*

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- Explain how ecological community composition changes over time as a result of primary and secondary succession.

  a) *Differentiate between primary and secondary succession. Provide an example of each to support your explanation.*

  ____

  ____

b) What types of species typically appear first during the pioneer stages of succession?

c) What types of species tend to be found only in climax communities?

- Explain the role of defence mechanisms in predation and competition.

a) List three defence "strategies" often employed by organisms to avoid predation.

1.

2.

3.

- Describe the factors that influence population growth (e.g. mortality, natality, immigration, emigration).

a) Complete the following table comparing the four main factors that influence population growth:

Factor	Definition	Does it increase or decrease

		population size?
Mortality		
Natality		
Immigration		
Emigration		

- Describe the growth of populations in terms of the mathematical relationship between carrying capacity, biotic potential, environmental resistance and the number of individuals in a population.

a) *Define each of the following terms.*

Carrying capacity

______ **Biotic potential**

______ **Environmental resistance**

- Calculate growth rate, per capita growth rate, population size and population density.

a) *Define each of the following methods of assessing population growth. Provide the mathematical formula used to determine each.*

	Definition	Formula
Population density		
Population size		
Growth rate		
Per capita growth rate		

b) *The population density of geese in a 1 km^2 area is 130 geese/km^2. Determine the number of geese found in this population.*

c) *In 2005, the population of moose in Red Deer was determined to be 335. By 2008, 78 of those moose had died, and 115 were born. Furthermore, 21 moose had left the area, and 13 joined.*

➢ Determine the change in population size between 2005 and 2008.

➢ Determine the growth rate between 2005 and 2008.

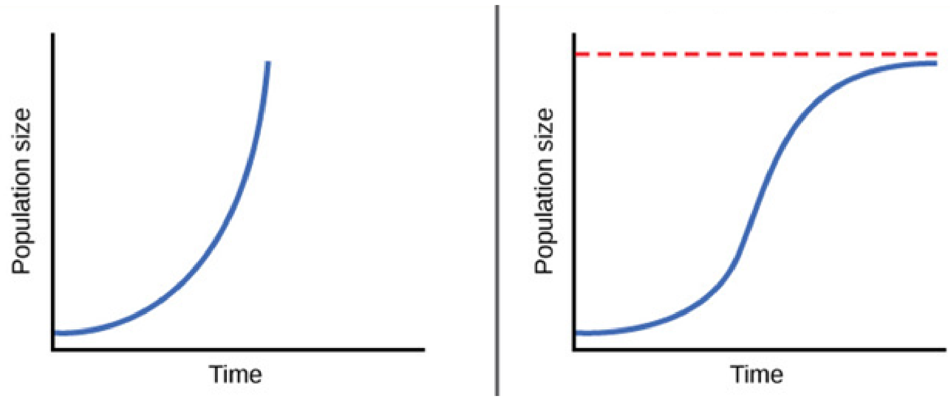
➢ Determine the per capita growth rate between 2005 and 2008.

➢ Is the population of moose in Red Deer growing or shrinking?

➢ List one density-dependent factor and one density-independent factor that could account for the change in moose population between 2005 and 2008.

- Explain the difference between logistic (S-shaped) and exponential (J-shaped) growth patterns.

  a) *Label the following growth patterns as either logistic or exponential. List some species that typically demonstrate each type of growth pattern.*



b) *Label the lag phase, log phase, and stationary phase on the graph above that demonstrates logistic growth.*

c) *Explain what causes the "levelling off" (stationary phase) of population size over time in logistic growth patterns.*

- Explain the characteristics of open and closed populations.

a) *Differentiate between an open population and a closed population.*

- Describe the characteristics and reproductive strategies of r-selected and K-selected organisms.

a) *Complete the following table comparing r-selected and K-selected life strategies:*

	r-selected	**K-selected**
Lifespan		
Reproductive age		
Body size		
Number of offspring		
Type of growth		